

Ph.D. exam on Measure and Integration Theory
Jan 1997

I. State the following theorems:

1. The monotone convergence theorem for nets in L^1
2. The Hahn decomposition theorem
3. The Egorov theorem
4. The Vitali convergence theorem
5. The Radon Nikodym theorem
6. The Fubini theorem
7. The representation of the dual of L^p .
8. The Lebesgue dominated convergence theorem

II. Prove one of the theorems 6 or 7.

III. Solve 5 of the following problems

In what follows, (X, Σ, μ) is a measure space.

- 1.) Let (f_n) be a sequence of real valued measurable functions on X and f a real measurable function on X .
Prove that (f_n) converges to f in μ -measure
($f_n \xrightarrow{\mu} f$) iff every subsequence (f_{n_i}) contains
a further subsequence $(f_{n_{i_j}})$ converging to f , μ -a.e.

- 2.) Let (f_n) be a sequence of positive, μ -integrable functions such that $\sum_n \int f_n d\mu < \infty$. For each $x \in X$ denote

$$L(x) = \sup \{ n; f_n(x) > 1/n \}$$

Prove that $L(x) < \infty$, μ -a.e.

- 3.) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a Lebesgue integrable function.
Prove that for every $\varepsilon > 0$, there is a bounded interval I such that $|\int_E f dx| < \varepsilon$ for every measurable set E with $E \cap I = \emptyset$.

- 4.) Let F be a Banach space and $\mathcal{F} \subseteq \mathcal{E}$ a sub σ -algebra. Prove the existence of the conditional expectation $E(f|\mathcal{F})$ for every μ -integrable function $f: X \rightarrow F$.
- 5.) Let $(X, \mathcal{F}, \mu), (Y, \mathcal{G}, \nu)$ be two real measure spaces. Prove the existence of the product measure $\mu \times \nu$ and that $|\mu \times \nu| = |\mu| \times |\nu|$.
- 6.) Let μ be the Lebesgue measure on $\mathbb{R}$ and f a Lebesgue integrable function on $\mathbb{R}$ such that $\int_0^x f d\mu = 0$ for every $x \in \mathbb{R}$ (if $x < 0$, $\int_0^x = -\int_x^0$). What can you say about f ?
- 7.) Let μ be the Lebesgue measure on $\mathbb{R}$ and $f_n = -\chi_{(n, \infty)}$, $n=1, 2, \dots$. Show that (f_n) is increasing but that the conclusion of the monotone convergence theorem is not true. Explain why.
- 8.) Let μ be the Lebesgue measure on $(0, 1)$. Each point $x \in (0, 1)$ has a binary expansion
$$x = \sum_{i=1}^{\infty} \frac{x_i}{2^i}, \text{ where } x_i = 0 \text{ or } 1.$$
 (The expansion is unique except a countable set, which we can throw out of the space).
- (i). Define $T_k(x) = x_k$. Sketch the graphs of T_1, T_2, T_3 .
- (ii) Let $\mathcal{F}_k = \sigma(T_1, T_2, \dots, T_k)$. Describe $\mathcal{F}_k$.
- (iii) Let $f \in L^1(\mu)$. Find a function g_1 which is $\mathcal{F}_1$ -measurable and satisfies
$$\int_A g_1 d\mu = \int_A f d\mu \text{ for } A \in \mathcal{F}_1$$
- (i.e. find $g_1 = E(f|\mathcal{F}_1)$). Find $g_2 = E(f|\mathcal{F}_2)$.